CCS2131 DATA STRUCTURES AND ALGORITHMS

MID TERM TEST

WEDNESDAY, 22 APRIL 2026

15:30 -17:00 HOURS

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Any any **TWO** questions from three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

Question 1

A logistics company has received the following eight package priority values:

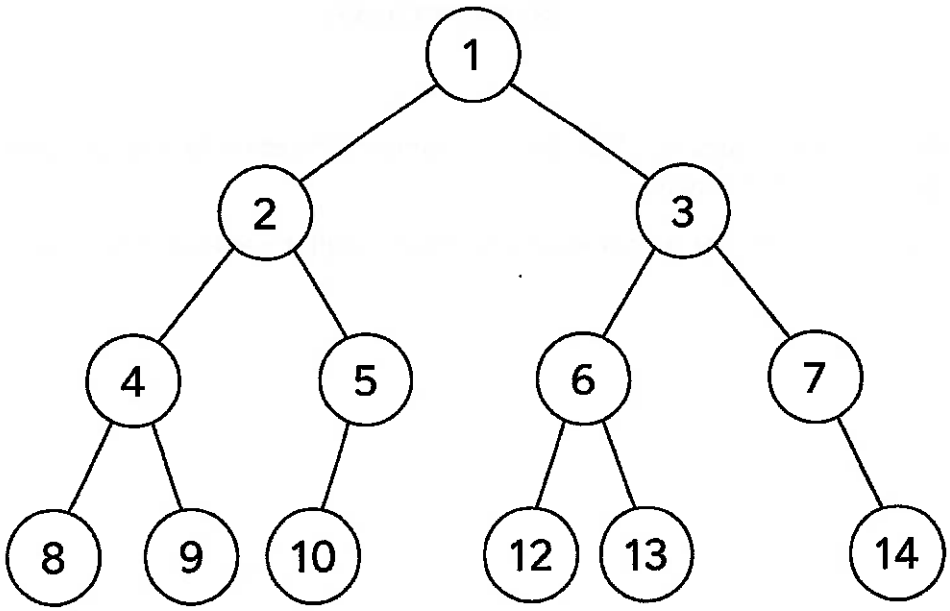
[17, 42, 68, 9, 73, 25, 51, 84]

The company wants to organize these values using a **maximum heap tree** so that the package with the highest priority is always processed first.

Create a maximum heap tree. Show all your workings. **(25 Marks)**

Question 2

A software company stores tasks in a binary tree structure so that each parent task may have sub-tasks under it as shown below. Before closing any main task, the system must first complete all of its sub-tasks. This means the system must process the left subtree first, then the right subtree, and finally the parent node.



Traverse the tree using postorder traversal **(25 Marks)**

Question 3

A small shop uses the following Python code to perform calculations on two numbers entered by the cashier:

1. ***a = 12***
2. ***b = 4***
3. ***sum = a + b***
4. ***product = a * b***
5. ***difference = a - b***
6. ***print(sum, product, difference)***

Calculate the complexity of the code below using the big-O notation. **(25 Marks)**